

Xanthomonas Cultivation Protocol

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To cultivate *Xanthomonas* under optimal laboratory conditions (30°C in NYG medium), generating healthy biomass for downstream experiments such as growth curve analysis and thermal stress assays.

Materials

- Activated *Xanthomonas* inoculum
- 6 × 125 mL Erlenmeyer flasks
- 50 mL NYG medium per flask
- Antibiotics (Kanamycin 20 µg/mL used)
- Refrigerated rotary shakers (30°C + stress temperature)
- Spectrophotometer
- NYG agar plates
- Sterile dilution tubes
- Pipettes
- Temperature monitor

Procedure

Plate Activation

1. Streak bacteria from glycerol stock onto NYG agar plate.
2. Add kanamycin if culturing mutants.
3. Incubate at 30°C for 48 hours.

Colonies should appear yellow and mucoid.

Liquid Culture

1. Pick a single isolated colony.
2. Inoculate into 5–50 mL NYG liquid medium.
3. Add kanamycin if required.
4. Incubate at 30°C, 200 rpm overnight (~12–16 h).

Measuring Culture Density

1. Blank spectrophotometer with NYG medium.
2. Measure OD600.
3. Log phase typically occurs around OD600 = 0.3–0.6.

This culture can now be used for:

- Phage infection
- Long-term storage
- Experimental assays

Glycerol Stock Preparation (Long-Term Storage)

1. Mix:
 - 800 µL overnight culture
 - 200 µL 50% glycerol
- Final glycerol concentration = 20%

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3. Store at -80°C